Interactive Web-Controlled LED System with Capacitive Touch using ESP32-S3

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This project aims to design and build an interactive LED system integrated into a 3D-printed enclosure. The system utilizes an ESP32-S3 microcontroller to manage five LEDs of different colors. It offers two control methods: wireless control via a local web server for individual and group LED toggling, and physical control using a capacitive touch sensor hidden beneath the enclosure's surface. The system demonstrated zero-latency response over the network and highly stable touch detection through the insulating layer, providing an effective model for embedded IoT applications.

Introduction

With the rapid advancement of the Internet of Things (IoT) and embedded systems, integrating digital control interfaces with physical interactions has become essential for developing smart products. This project leverages the processing and connectivity capabilities of the ESP32-S3 to build an integrated electronic circuit that controls lighting. The ESP32-S3 was specifically chosen for its built-in Wi-Fi support and high efficiency in processing capacitive touch signals without the need for complex external hardware, making it the optimal choice for this engineering project.

Hardware Design

The electronic circuit was designed to be stable, safe, and suitable for the internal dimensions of the 3D-printed box. The hardware architecture consists of:

- **Microcontroller (ESP32-S3 Dev Module):** Acts as the brain of the system, connected to a breakout board for easier and more secure wiring.
- **LED Indicators:** Five LEDs (White, Yellow, Red, Green, Blue) connected to digital output pins (13, 12, 14, 11, 10) respectively.
- **Touch Sensor System:** A capacitive sensing field was created using highly conductive Nickel strips connected to pin `IO4`. They are attached to the inner roof of the box, enabling highly efficient touch detection through the plastic surface.

- **Common Ground:** To ensure circuit stability, a breadboard was used to unify the ground (GND) line for all five LEDs and connect it to the main GND pin on the microcontroller, while strictly observing LED polarity.

Software Implementation

The system was programmed using the Arduino IDE, with the logic divided into two main sections:

- **Local Web Server:** The microcontroller was programmed to connect to a Wi-Fi network and host a lightweight HTML page. This page sends HTTP requests to the ESP32-S3 based on user input to toggle specific LEDs or all of them simultaneously, keeping the LED states perfectly synchronized.
- **Touch Algorithm:** The microcontroller continuously reads the capacitance values and categorizes physical touches into three distinct actions:
    - **Short Press:** Cycles to the next LED color and turns off the previous one.
    - **Double Press:** Adds a new color while keeping the currently active LEDs on.
    - **Long Press:** Resets the system and turns off all LEDs.

Challenges & Troubleshooting

During the development and testing phases, two main engineering challenges were encountered and successfully resolved:

- **Ghost Touches (False Triggers):** Initially, the sensor was overly sensitive, causing random LED activations. This was solved by analyzing raw capacitance readings via the Serial Monitor. The idle reading (~25,600) and the actual touch reading (~28,000) were identified. The touch threshold was then programmatically calibrated to `27000`, eliminating false triggers and ensuring precise touch detection.

- **Insulation in the Touch System:** Using aluminum foil with adhesive tape created an insulating barrier that weakened touch detection from the top of the box. This was resolved by discarding the foil and using Nickel strips to concentrate the electromagnetic field. This modification allowed the microcontroller to effectively and accurately detect touches through up to 5mm of solid plastic.

Conclusion & Future Work

The project successfully met its design objectives, delivering a stable, integrated system with precise web control and zero-latency touch response, perfectly housed within a custom 3D-printed physical design.

Future Enhancements:

1. Connecting the system to cloud servers (e.g., Blynk platform) to allow global remote control via a smartphone application.
2. Adding a small OLED display to show the local IP address and the current status of the LEDs.
3. Integrating a Light Dependent Resistor (LDR) so the system can automatically adjust LED brightness based on ambient room lighting.